

Fixed IFEval-Like Loading Simulation

This simulation is designed to test whether the product MAP estimator gives factor and parameter recovery competitive with Viroli-style probit Gibbs samplers, while being substantially faster in larger item and factor settings.

Scientific Question

The target question is:

Holding the population model fixed, does recovery improve as the number of items loading on each factor increases, and how does product MAP compare with Viroli Gibbs under the same data-generating model?

The simulation therefore fixes the population parameters within each design cell and generates a new binary response dataset in each replication.

Data-Generating Model

For observation i and item j ,

```
X_ij = 1{Z_ij > 0}
Z_ij = alpha_j + f_i' lambda_j + epsilon_ij
epsilon_ij ~ N(0, 1)
```

The latent factors are independent marginal mixtures:

```
f_ih ~ sum_g pi_hg N(mu_hg, sigma_hg^2)
```

The mixture parameters are generated with MIXTURE_PARAM_MODE=viroli_smoke. For `sep = 1`, the raw component parameters are:

- $G_h = 2$: weights (0.5, 0.5), means (-1, 1), sds (0.55, 0.85).
- $G_h = 3$: weights (0.3, 0.4, 0.3), means (-1.35, 0, 1.35), sds (0.45, 0.65, 0.45).

Each factor coordinate is then standardized before generating responses, so the latent factor scale is comparable across settings.

Fixed Loading Design

Only one loading design is used in the main simulation:

- IFEval-like unbalanced item blocks.
- The smallest primary-loading block has at least 30 items.
- Nonzero loading magnitudes are sampled from `Uniform(2, 3)`.
- Cross-loading probability is 0.05.
- Cross-loading signs are random.
- Item intercepts use the IFEval-like intercept design.

For each H , a master loading matrix is generated at $p_{\max} = 2000$. Smaller p settings use nested block-wise subsets of this master matrix. Thus increasing p increases item support without redrawing the population loading matrix.

Simulation Grid

The main grid is:

- n in $\{100, 200, 400\}$
- p in $\{500, 1000, 1500, 2000\}$ for product MAP
- p in $\{500, 1000\}$ for the Gibbs baselines
- H in $\{5, 10\}$
- $G_h = 2$ for every factor, or $G_h = 3$ for every factor
- 25 replications per design cell

The alternating or mixed G_h setting is intentionally excluded to keep the interpretation focused.

Methods

Three methods are compared.

1. Product MAP:
 - EM-SVD probit signal pretraining.
 - Riemannian rotation toward independent marginal mixtures.
 - MAP refinement with lasso-penalized loading updates.
 - Uses 18 internal workers.
2. Viroli Laplace Gibbs:
 - Probit-augmented independent-mixture Gibbs sampler.
 - Laplace loading prior with penalty 10.
 - Uses 4 internal workers.
 - Run for 2000 iterations with 1000 burn-in draws.
3. Viroli Gaussian Gibbs:
 - Same independent-mixture Gibbs sampler.
 - Diffuse Gaussian loading prior.
 - Uses 4 internal workers.
 - Run for 2000 iterations with 1000 burn-in draws.

Replications are run serially at the launcher level. Parallelization occurs within each fitted method.

Metrics

Each replication records:

- factor score RMSE after alignment
- raw factor score RMSE
- loading RMSE
- intercept RMSE
- marginal mixture mean RMSE
- marginal mixture variance RMSE
- marginal mixture weight RMSE
- probability RMSE
- stage-one signal and subspace diagnostics for product MAP
- end-to-end runtime in seconds
- Gibbs effective sample size summaries when available

The output also records the fixed DGP seeds:

- `loading_parameter_seed`
- `mixture_parameter_seed`
- `data_seed`

This makes it possible to verify that population parameters are fixed across replications while the sampled datasets change.

Reproduction

From the repository root:

```
Rscript scripts/sample_size/plot_fixed_ifeval_lambda_heatmaps.R
Rscript scripts/sample_size/run_fixed_ifeval_lambda_simulation.R
Rscript scripts/sample_size/plot_fixed_ifeval_lambda_progress.R
```

For a short smoke run:

```
REP_VALUES=1,2 \
N_VALUES=100 \
P_VALUES_PRODUCT=500 \
P_VALUES_GIBBS=500 \
H_VALUES=5 \
G_CONFIG_TYPES=all2 \
  RUN_LABEL=fixed_ifeval_lambda_min30_u2_3_cp0_05_h5_h10_smoke \
Rscript scripts/sample_size/run_fixed_ifeval_lambda_simulation.R
```

Main output root:

```
results/full/fixed_ifeval_lambda_min30_u2_3_cp0_05_h5_h10/
```

Within that directory:

- `comparison_results.csv`: combined raw results after phase checkpoints.
- `chunks/`: chunk-level logs and checkpoint result files.

Interim and final plots are written by:

```
scripts/sample_size/plot_fixed_ifeval_lambda_progress.R
```

Deterministic DGP artifacts use the same run label:

```
fixed_ifeval_lambda_min30_u2_3_cp0_05_h5_h10
```

- Heatmaps live under `results/selected_plots/sample_size/`.
- Loading matrices live under `results/selected_tables/sample_size/`.